

Journal Paper 4 Summary (25 points total)

Paper Title	Infovis and Statistical Graphics: Different Goals, Different Looks
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1. What do you think the paper is about in layman's terms? What are the main arguments presented and any recommendations/suggestions about the tension and need for collaboration between practitioners in the fields of statistical graphics and information visualization? Do you agree with the recommendations/suggestions? Why or why not? [5 points]

Focus of this question
• This question encourages you to evaluate the arguments, evidence, assumptions, and conclusions about key issues (i.e. think critically about the paper) • This question encourages you to develop your own knowledge, comprehension and conceptual understanding and to connect, synthesize, and/or transform your ideas into a new form (i.e. be a creative thinker and contribute your ideas and thoughts) • This question challenges you to consider diverse perspectives (gender, political, ethnic, racial, etc) during class or in assignments

Write your answer below:

This paper is about the gap between statistical modeling and information visualization, and how to connect the two to work cohesively with each other. The purpose of the paper is the “start a conversation between practitioners in statistical graphs and [information visualization]”.

The main arguments are as follows: discrepancy between visual appeal and actual correct use and presentation of statistical graphs, insufficient understanding of the goals of data display, and distinction between exploratory and presentation graphics.

The recommendations about collaborations can be broken into two main goals: discovery and communication. For discovery goals, the authors highlighted giving a qualitative sense to the dataset by confirming known results and highlighting distinct patterns, conveying the scale and complexity of a dataset to paint an entire picture of the story, and to explore the

data by utilizing flexible displays to uncover unexpected aspects of the data. For the communication goals, the authors highlighted displaying information in an understandable way, telling a compelling story, and attracting attention and simulating interest.

I agree with all these suggestions. I think that to display information accurately and transparently to an audience that may not have the necessary background is crucial to gain their trust, so they can make their own well-informed decisions, whether that be with words or actions. I believe that presenting information in an accurate way not only helps the audience/stakeholder, but also helps the data scientists gain necessary background info into the data they are collecting, helping them deduce and uncover trends and patterns that are not explicitly read in the data.

2. The authors present a discussion of the key differences between the two approaches of information visualization and statistical graphics. Think about these differences, mention and discuss four real-world applications of one key difference between information visualization and statistical graphics. [5 points]

Focus of this question

- This question encourages you to connect your learning to “real world” issues or life experiences and consider diverse perspectives for the application of concepts in the paper to the real-world
- This question encourages you to develop your own knowledge, comprehension and conceptual understanding and to connect, synthesize, and/or transform your ideas into a new form (i.e. be a creative thinker and contribute your ideas and thoughts)

Write your answer below:

One key difference between information visualization and statistical graphics essentially boils down to how a graph functions vs. how it looks. For statisticians creating statistical graphs, they are more concerned with how accurately the data is being displayed rather than how it looks. On the other end, information visualization is more about how aesthetic a graphic can look rather than on displaying the data in a purely functional way.

Real world applications:

Pie charts are used very commonly when displaying data, even if they are not the most accurate. Pie charts are popular because they allow us to compare parts-to-whole in a relatively aesthetic way. However, they are not practical at all, because judging a pie chart and eyeballing how close the slices are in size to each other is quite difficult to do without having to explicitly state how big each slice is. I have personally seen pie charts used quite often, and while you can get a good picture when things are extreme, they fail when the data is more closely related.

When displaying data, while simple is best, we are still creatures that appreciate pretty things (ooh, shiny!). Creating simple line charts and bar charts are more than enough to show categorical differences, change over time, etc. yet if the graph does not evoke any emotion or compel you towards a certain action, is there even a point to displaying accurate information? Yes, the information is accurate, but it will not persuade you one way or another. Where information visualization hits, its stride is appealing to our emotions and getting a response out of us. If you are trying to persuade a stakeholder to side with you, aesthetics matter.

One example of an aesthetic visualization that does not have statistical value are word clouds that use size as a representative of importance. The larger a word, the more

important it is. This is pleasing to the eye but doesn't really do much outside of that. Size is a good indicator, but not accurately. It's the same problem that occurs with pie charts: the closer in size two words are the harder it is to tell which one is bigger.

Finally, there are many ways to display statistical information, yet parsing this data and finding/analyzing these patterns takes experience and knowledge of statistics. Since information visualization is more aesthetically pleasing, sometimes information graphics can miss the mark on what they are trying to represent, purely because the person/team creating it do not actually know how statistics work.

3. Choose one of the “Communication goals” (discussed on pages 7 – 9 of the paper) and discuss at least three (3) implications of your chosen communication goal for the visual display of quantitative information not mentioned in the paper. [15 points]

Focus of this question
• This question encourages you to connect your learning to “real world” issues or life experiences and consider diverse perspectives for the application of concepts in the paper to the real-world • This question encourages you to reflect on what you are learning • This question encourages you to contribute your ideas and thoughts • This question challenges you to consider diverse perspectives (gender, political, ethnic, racial, etc.) during class or in assignments • This question challenges you to develop and present your own knowledge, comprehension, and conceptual understanding

Write your answer below:

The communication goal that I will choose is “Telling a story”.

1. Storytelling is a big component of good information design because with this data, we want to convince the stakeholder that our point is the one to stand with. Using data to back up our claim is integral in persuading the stakeholder to consider our perspective.
2. Everybody loves to hear about stories, they have driven societies since the beginning of our species (not fact-checked). There’s a reason that the same folk-stories, legends, myths, etc. are repeated often and looked back on whether to teach a lesson or simply for enjoyment. Using data to do both is an indicator of a good visual/graphic.
3. We are humans trying to convince other humans that our story matters. Stakeholders need evidence to conclude if our stance is really the right one to choose. What better way than to do it with data and with *correct* data? Turning numbers and statistics into a deeply personal thing by changing how they are being presented and appealing to the stakeholder’s emotions is a skill not many people have, and is integral in convincing your stakeholder to appeal to your opinion and purposes.